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RESEARCH ARTICLE

Dual Transformers With Latent Amplification for Multivariate Time Series Anomaly Detection

YEJI CHOI^{1,2}, KWANGHOON SOHN¹, (Senior Member, IEEE),
AND IG-JAE KIM², (Member, IEEE)

¹School of Electrical and Electronic Engineering, Yonsei University, Seoul 03722, South Korea

²Artificial Intelligence and Robotics Institute, Korea Institute of Science and Technology, Seoul 02792, South Korea

Corresponding author: Ig-Jae Kim (drjay@kist.re.kr)

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ABSTRACT Anomaly detection in multivariate time series is crucial for applications such as industrial monitoring, cybersecurity, and healthcare. Transformer-based reconstruction methods have recently shown strong performance but often suffer from overgeneralization, where anomalies are reconstructed too accurately, thereby reducing the separability between normal and abnormal patterns. Prior works have attempted to mitigate this by incorporating two-stage frameworks or external memory modules to explicitly store normal patterns and amplify deviations from abnormal patterns. However, such approaches increase model complexity and incur additional computational overhead. In this paper, we propose Dual Transformers with Latent Amplification (DT-LA), a novel framework designed to mitigate overgeneralization within a unified architecture. The core idea of DT-LA is to enhance anomaly separability by jointly leveraging both input and latent space reconstructions, rather than merely improving reconstruction fidelity. In particular, we propose the Modified Reverse Huber (MRH) loss that amplifies meaningful deviations in the latent space by applying inverse scaling. It allows the model to retain informative discrepancies that would otherwise be suppressed, thereby improving its ability to detect subtle anomalies. Second, we incorporate sparse self-attention with entropy-based regularization to capture essential inter-sensor relationships and suppress redundancy. Third, we refine the anomaly scoring process using a scaled-softmax function, which balances relative and absolute deviations to reduce softmax-induced bias. Extensive experiments on four benchmark datasets (SMAP, MSL, PSM, and SMD) demonstrate that DT-LA achieves state-of-the-art performance, with F1-scores of 97.02% on SMAP and 98.42% on PSM, highlighting its robustness and practical competitiveness as a single-stage framework.

INDEX TERMS Anomaly detection, multivariate time series, sparse self-attention, transformer.

I. INTRODUCTION

Anomaly detection in multivariate time series aims to identify anomalous time points where the correlations among variables deviate from normal patterns while also considering temporal trends. This task is crucial in various applications, such as industrial monitoring, cybersecurity, and healthcare systems, where detecting irregularities is essential for maintaining system reliability and safety.

Traditional machine learning techniques such as one-class SVM (OC-SVM) [1], Isolation Forest [2], and Local Outlier Factor (LOF) [3] provide foundational baselines but struggle

to model complex temporal and inter-sensor dependencies. In contrast, recent deep learning-based models [4], [5], [6], [7], [8], [9], [10], especially based on LSTM [11] and Transformer [12], have advanced performance by effectively capturing sequential and global context. Generative [13], [14] and graph-based methods [15], [16] further enrich representations through adversarial training and structured dependency modeling.

Among them, reconstruction-based methods [6], [7], [8], [9], [10], [17] represent a mainstream paradigm under weakly supervised settings, where models are trained exclusively on normal data and anomalies are identified based on reconstruction errors. Recent studies have further extended this idea to the latent space [18], reconstructing latent

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representations to capture more abstract patterns using additional autoencoder structures [19], [20] or statistical modeling techniques [21], [22]. These methods have demonstrated strong performance across diverse datasets, largely due to the enhanced expressiveness and generalization capabilities of modern autoencoding architectures.

However, such models inherently suffer from the overgeneralization problem [23], [24], wherein even anomalous inputs are reconstructed with high fidelity, thereby reducing the separability between normal and abnormal patterns. For instance, in sensor-driven system monitoring scenarios, minor sensor failures or subtle drifts may be overly well reconstructed due to over-expressive decoders [10] or spurious inter-sensor correlations [9]. As a result, these systems may fail to trigger timely alarms in real-world applications, potentially leading to critical safety issues or substantial operational losses.

To mitigate this challenge, recent models such as TranAD [9] and MEMTO [10] have advanced reconstruction-based anomaly detection by integrating adversarial training, self-conditioning, or memory modules. However, these methods rely on complex multi-phase optimization or external memory mechanisms (e.g., K-means-based prototype storage), which increase computational overhead and complicate training. Moreover, few methods tackle the overgeneralization problem in a single-stage setting that avoids such complexities. These limitations highlight the need for a lightweight, single-stage framework that amplifies informative reconstruction errors without external modules—enhancing both robustness and real-time deployability.

In this paper, we propose Dual Transformers with Latent Amplification (DT-LA), a novel framework designed to mitigate overgeneralization in time-series anomaly detection within a unified single-stage architecture. The core idea of DT-LA is to enhance anomaly separability by jointly leveraging both input and latent space reconstructions, rather than merely improving reconstruction fidelity. To this end, we introduce a Modified Reverse Huber (MRH) loss, which acts as a hybrid of L2 and inverse loss: small deviations are treated quadratically to maintain accurate reconstructions, whereas larger deviations are penalized less aggressively. This enables the model to retain anomaly-related signals that would otherwise be smoothed out. Additionally, we incorporate a sparse self-attention mechanism to capture essential inter-sensor dependencies, which is an essential capability for handling noisy or weakly correlated multivariate time series. By applying entropy-based regularization to the attention weights, the model is encouraged to focus on the most informative relationships while suppressing redundant or noisy ones. This design eliminates the need for predefined sensor associations and does not impose significant computational overhead. Finally, we refine the anomaly scoring process by introducing a scaled-softmax function that balances relative and absolute deviations, alleviating softmax-induced distortion. This refined criterion prevents

unintended emphasizing of minor fluctuations in normal data, thereby enhancing detection precision and reducing false positives.

Extensive experiments on four widely used benchmark datasets—SMAP, MSL, PSM, and SMD—demonstrate that DT-LA consistently outperforms state-of-the-art baselines, achieving F1-scores of 97.02% on SMAP and 98.42% on PSM. Our model shows balanced improvements in both precision and recall, effectively detecting subtle anomalies while minimizing false positives. These results highlight DT-LA's efficiency, scalability, and practical applicability in real-world anomaly detection scenarios.

In summary, this paper addresses the overgeneralization problem in reconstruction-based approaches for time series anomaly detection by proposing a novel framework that integrates dual-space reconstruction. The main contributions of this work are as follows:

- 1) We propose Dual Transformers with Latent Amplification (DT-LA), a unified framework that jointly uses input and latent reconstruction errors to improve anomaly separability while mitigating overgeneralization.
- 2) We introduce the Modified Reverse Huber (MRH) loss, which amplifies meaningful latent deviations through inverse scaling, allowing the model to preserve informative signals and detect subtle anomalies.
- 3) We propose a sparse self-attention mechanism with entropy-based regularization to capture essential inter-sensor dependencies without relying on predefined structures, enhancing computational efficiency.
- 4) We propose a refined anomaly scoring function based on scaled-softmax, which effectively balances relative and absolute reconstruction deviations and mitigates softmax-induced distortions, reducing false positives.
- 5) Our method demonstrates consistently superior performance on four benchmark datasets, with balanced improvements in F1-score, precision, and recall. Its single-stage, memory-free design ensures both efficiency and practicality for real-world applications.

The rest of the paper is organized as follows. Section II provides a brief review of existing methods for anomaly detection in multivariate time series. Section III presents an overview of our proposed framework and details the three key components. Section IV discusses the experimental results, ablation studies, and analysis. Finally, Section V concludes the paper.

II. RELATED WORK

We categorize existing multivariate time-series anomaly detection work into two primary streams: (1) model-based approaches that use autoencoders, RNNs, VAEs, GANs, or graph structures to learn normal patterns, and (2) attention-based approaches, particularly Transformer-based methods,

which are widely used for their ability to capture long-range temporal and inter-sensor dependencies.

A. MODEL-BASED APPROACHES

Early methods under weak supervision focus on defining statistical or machine learning boundaries for normal behavior. Traditional algorithms such as One-Class SVM [1] and Isolation Forest [2] learn compact representations of normal data, while density-based methods like KNN [25], Gravitational Anomaly [26], and LOF [3] detect outliers based on local neighborhood properties. Rule-based classifiers, including decision trees [2], [27], offer simple interpretability. However, these traditional methods generally struggle to scale to high-dimensional and complex time series data, limiting their effectiveness in modern industrial applications.

To overcome these limitations, deep learning techniques have become the dominant paradigm, leading to significant improvements in anomaly detection performance. Autoencoder-based methods such as VAEs [28], RNNs [29], and LSTM-VAEs [5] enable effective representation learning from normal training sequences, while OmniAnomaly [6] extends the LSTM-VAE framework with normalizing flows and probabilistic reconstruction. InterFusion [7] further captures complex dependencies using a hierarchical VAE. GAN-based models [13], [14], [30], [31] leverage adversarial training to enhance reconstruction fidelity. Meanwhile, graph-based approaches such as DVGCRN [15] and MTAD-GAT [16] explicitly model sensor relationships by embedding predefined dependency structures into the learning process.

In parallel to architectural advances, recent studies have adopted contrastive learning to improve representations under limited or noisy supervision. Methods like PCRTA [32] and MulGad [33] boost anomaly sensitivity via temporal and multi-scale embedding strategies, but often rely on complex augmentations and sampling schemes.

While several prior works incorporate both input and latent reconstruction errors to enhance overall reconstruction fidelity [19], [21], their goal is typically to minimize all deviations. In contrast, our method uses latent reconstruction to selectively preserve meaningful anomalies, enhancing separability without multi-stage training. This targeted amplification of informative deviations distinguishes DT-LA from existing approaches that may suppress subtle anomaly cues.

B. ATTENTION-BASED APPROACHES

Inspired by the success of attention-based methods in domains such as natural language processing [34], audio [35], and video analysis [36], Transformer architectures have recently demonstrated strong performance in time series anomaly detection. Their self-attention mechanism enables global context modeling, which is crucial for capturing delayed or subtle anomalies in multivariate settings.

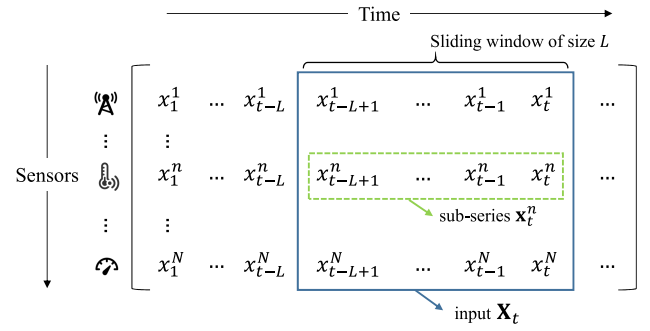


FIGURE 1. Illustration of multivariate time series with a sliding window of size L .

TABLE 1. Notations.

X_t	Multivariate input time series at time t
x_t^n	Sub-series of the n -th variable at time t with length L
L	Sliding window length
E_{pri}	Primary transformer encoder
E_{sec}	Secondary transformer encoder
z_t	Latent representation from E_{pri}
$\hat{z}_t$	Latent representation from E_{sec}
$\hat{X}_t$	Reconstructed output
M_t	Self-attention map at time t
λ	Weight for sparse self-attention regularization loss
δ	Threshold parameter in Modified Reverse Huber (MRH) loss

Anomaly Transformer [8] leverages attention mechanisms that capture association discrepancies to differentiate anomalous patterns from normal ones. Building on this, DCdetector [37] employs contrastive learning to enhance the separation of anomalous behaviors through attention, while DUMA [38] introduces max-mask self-attention to effectively model sensor-level dependencies. TranAD [9] employs a Transformer-based network with adversarial training and self-conditioning in a two-phase framework to reinforce the deviation between reconstructed and actual inputs. Meanwhile, MEMTO [10] utilizes memory modules that store normal prototypes via K-means clustering and integrates memory-guided attention with a dual-scoring strategy to improve anomaly localization.

While MEMTO and DCdetector improve attention via memory augmentation and contrastive objectives, they rely heavily on explicit storage or pairwise learning schemes, which may be sensitive to augmentation quality or suffer from memory saturation. In contrast, our proposed DT-LA adopts a dual-attention framework that naturally emerges from reconstructing both input and latent spaces, without requiring external memory or contrastive pairs.

Recent extensions to the Transformer framework integrate graph structures into the attention mechanism, enabling more expressive modeling of inter-sensor dependencies. TopoGDN [39] combines graph neural networks with topological attention to capture multi-scale sensor dependencies in a temporally-aware fashion. Similarly, the GAT+Informer

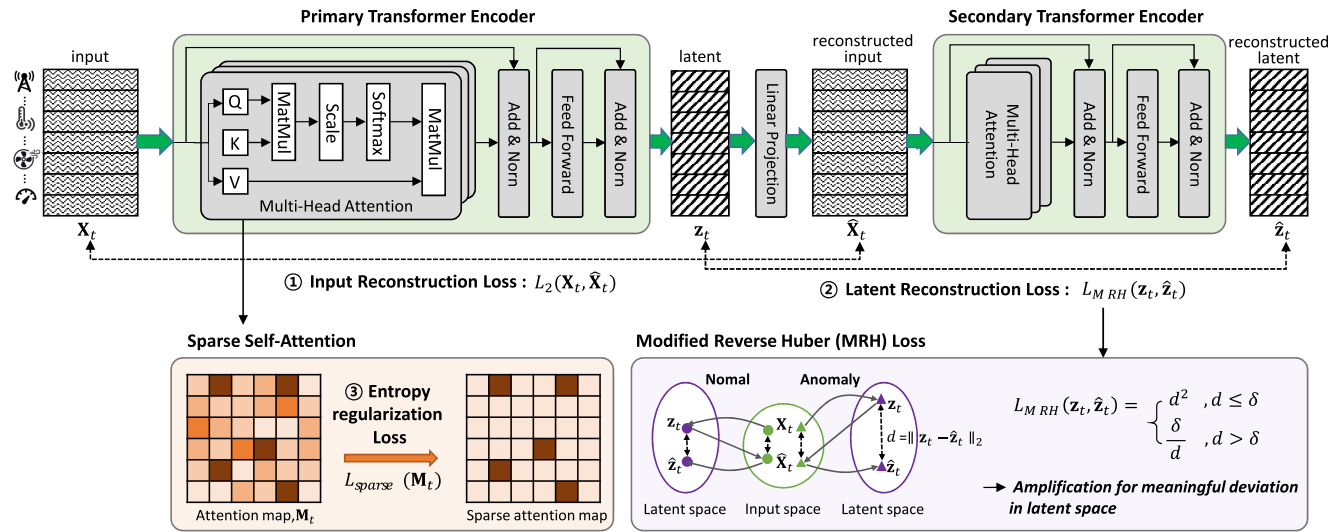


FIGURE 2. Illustration of the proposed Dual Transformers with Latent Amplification (DT-LA) framework. The framework includes a primary and secondary Transformer encoder to reconstruct input and latent spaces while amplifying meaningful latent space deviations via the Modified Reverse Huber (MRH) loss. Sparse self-attention in the Primary Encoder highlights essential sensor relationships and reduces redundant dependencies using entropy-based regularization. The entire network is trained end-to-end in a unified single-stage framework.

hybrid model [40] enhances long-sequence forecasting and anomaly detection by fusing global attention with graph-based spatial priors. Unlike these methods, DT-LA does not assume any prior sensor topology. Instead, it implicitly learns inter-sensor relationships via a sparse self-attention mechanism regularized by entropy, allowing for adaptive sparsity and interpretability without relying on external graph definitions.

III. PROPOSED METHOD

A. OVERVIEW

Fig. 1 illustrates an example of a multivariate time series with a sliding window, and Table 1 summarizes the notations used throughout this paper. In a weakly supervised setting for anomaly detection, reconstruction-based approaches assume that all training data are normal and aim to reconstruct the input accurately. Conventional methods typically identify anomalies by assessing whether the discrepancy between the original input and its reconstructed output exceeds a predefined threshold. However, these methods often suffer from overgeneralization, where the model reconstructs anomalous data too well, reducing its ability to distinguish between normal and abnormal instances. To address this issue, we propose Dual Transformers with Latent Amplification (DT-LA), a novel single-stage framework that performs reconstruction in both input and latent spaces while amplifying latent deviations to improve anomaly sensitivity.

Formally, for a given multivariate time series with N variables, we denote the input as $\mathbf{X}_t = \{\mathbf{x}_t^1, \dots, \mathbf{x}_t^N\}$, where each sub-series $\mathbf{x}_t^n$ represents the observation of variable n over the sliding window of size L , i.e.,

$$\mathbf{x}_t^n = [x_{t-L+1}^n, x_{t-L+2}^n, \dots, x_t^n]. \quad (1)$$

As shown in Fig. 2, DT-LA employs two Transformer-based encoders: one reconstructs the input sequence, and the other re-encodes the reconstructed output into the latent space. Both encoders are trained jointly using a total loss that combines reconstruction objectives from both spaces. Sparse self-attention is employed during input reconstruction to capture essential inter-sensor dependencies, while the Modified Reverse Huber (MRH) loss is applied in the latent space to selectively amplify informative deviations. By integrating deviations from both spaces, DT-LA produces a refined anomaly score at each time step t , enabling robust and interpretable anomaly detection. The following subsections detail each component of the proposed method.

B. DUAL TRANSFORMERS

The Dual Transformers framework consists of two Transformer-based encoders and a linear projection layer: the Primary Transformer Encoder and the Secondary Transformer Encoder, which operate sequentially. The linear projection layer connects the two encoders, enabling the transformation of latent features and reconstruction of the input data.

As shown in top of the Fig. 2, the Primary Transformer Encoder ($\mathbf{E}_{\text{pri}}$) maps the input time series into the latent space, with the goal of reconstructing the input data through the subsequent linear projection layer. Given an input sequence $\mathbf{X}_t \in \mathbb{R}^{N \times L}$, a latent representation $\mathbf{z}_t$ is defined as:

$$\mathbf{z}_t = \mathbf{E}_{\text{pri}}(\mathbf{X}_t), \quad (2)$$

where $\mathbf{z}_t \in \mathbb{R}^{d_l \times L}$ with a latent dimension d_l . A simple linear projection layer then reconstructs the original

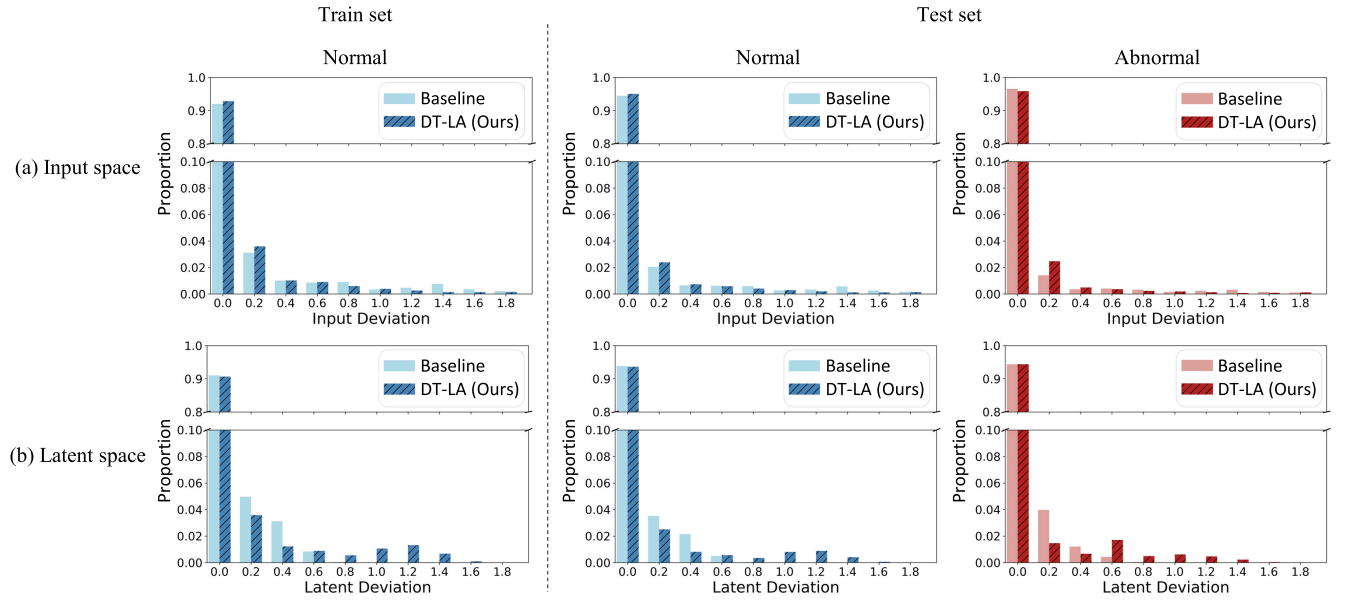


FIGURE 3. Proportional histograms of reconstruction deviation in (a) input space and (b) latent space on the SMAP dataset, comparing the Baseline and the proposed DT-LA. The Baseline is trained using L2 loss in both spaces, whereas DT-LA uses L2 loss in the input space and MRH loss in the latent space. DT-LA exhibits broader tails in the latent space, preserving informative deviations rather than suppressing them.

input, defined as:

$$\hat{\mathbf{X}}_t = \mathbf{W}\mathbf{z}_t + \mathbf{b}, \quad (3)$$

where $\mathbf{W} \in \mathbb{R}^{N \times d_l}$ and $\mathbf{b} \in \mathbb{R}^{N \times L}$ are trainable parameters for the transformation. Instead of employing a complex decoder, we adopt a simple linear projection layer to enforce a stronger bottleneck between the latent space and the reconstruction. This architectural choice is grounded in our goal of mitigating the risk of overgeneralization, rather than achieving high-fidelity reconstruction [10].

To further leverage latent reconstruction errors, the Secondary Transformer Encoder remaps the reconstructed output into a secondary latent representation $\hat{\mathbf{z}}_t \in \mathbb{R}^{d_l \times L}$, defined as:

$$\hat{\mathbf{z}}_t = \mathbf{E}_{\text{sec}}(\hat{\mathbf{X}}_t). \quad (4)$$

This additional encoding step allows the model to capture higher-dimensional reconstruction information in the latent space, improving its ability to model complex dependencies.

For optimization purposes, we use the typical reconstruction loss in the input space, which is the L2 loss, defined as:

$$\mathcal{L}_{\text{input}} = \|\mathbf{X}_t - \hat{\mathbf{X}}_t\|_2^2. \quad (5)$$

For the latent space reconstruction, however, our goal is not only to achieve accurate reconstruction in the input space but also to amplify errors when meaningful deviations occur during the input reconstruction process. To address this, we apply the MRH loss for latent reconstruction, a specialized loss function that we introduce in the next subsection, using L2 loss only for input reconstruction.

C. MODIFIED REVERSE HUBER LOSS

To overcome overgeneralization, we introduce the Modified Reverse Huber (MRH) loss, which amplifies deviations in the latent space. Inspired by the original Huber loss that balances L2 and L1 behaviors depending on error magnitude, MRH loss assigns different penalty forms based on a transition threshold.

The key feature of MRH loss is its inverse scaling behavior: unlike standard loss functions that uniformly increase the penalty along deviations, MRH loss reduces the penalty beyond a certain threshold. This encourages the model to preserve meaningful deviations rather than suppressing them, and is formally defined as:

$$\mathcal{L}_{\text{MRH}}(d) = \begin{cases} d^2, & \text{if } d \leq \delta, \\ \frac{\delta}{d}, & \text{otherwise,} \end{cases} \quad (6)$$

where $d = \|\mathbf{z}_t - \hat{\mathbf{z}}_t\|_2$ denotes the latent deviation, and δ is the transition threshold, which is empirically determined on the validation set for each dataset.

This design allows MRH loss to behave like L2 loss for small deviations (i.e., $d \leq \delta$), promoting accurate reconstruction, while applying an inverse scaling for larger deviations ($d > \delta$), where the loss decreases as the discrepancy increases. As a result, the model is discouraged from minimizing large latent deviations, enabling it to retain anomaly-related signals and thereby reducing overgeneralization. Formally, the gradient of MRH loss peaks near the transition threshold ($d \approx \delta$), so deviations just beyond this point receive the strongest learning signal. As d increases further, the gradient gradually diminishes due to the inverse

scaling mechanism, thereby reducing the penalty for large deviations.

In Fig. 3, we present proportional histograms of reconstruction deviations in both the input and latent spaces to illustrate how the proposed MRH loss affects the distribution of errors. While the Baseline is trained with L2 loss in both spaces, DT-LA applies L2 loss in the input space and MRH loss in the latent space. In (a), the input space, both models show similar deviation distributions concentrated near zero, indicating that DT-LA maintains stable input reconstruction despite the modified latent loss. In (b), the latent space, however, a clear divergence emerges. The baseline model produces sharply peaked deviations around zero, indicating overgeneralization. In contrast, DT-LA exhibits a broader distribution with heavier tails, reflecting a higher proportion of samples with large deviations. Notably, for abnormal samples, while the baseline shows near-zero frequency for deviations larger than 0.8—failing to preserve informative signals—our method retains a non-negligible proportion of such high deviations, indicating better preservation of subtle anomalies. This tail expansion demonstrates that MRH loss deliberately reduces penalization for large deviations and instead preserves potentially informative discrepancies in the latent space. This effect is not due to poor training, as the latent loss of DT-LA converges stably, but instead reflects the deliberate design of the MRH loss. This validates its core objective: enhancing anomaly separability by avoiding the over-suppression of abnormal signals and effectively mitigating the overgeneralization problem.

D. SPARSE SELF-ATTENTION

To capture essential sensor correlation, we incorporate a sparse self-attention mechanism into the Primary Transformer Encoder. This allows the model to focus on key interdependencies among sensors, discarding irrelevant ones.

The self-attention mechanism computes pairwise attention, encoding relationships between sensor variables across the time window. Given the hidden layer output $\mathbf{f}_t \in \mathbb{R}^{d_l \times L}$ in the encoder, the queries, keys, and values are obtained through linear projections as follows:

$$\mathbf{Q}_t = \mathbf{f}_t^T \mathbf{W}^q, \quad \mathbf{K}_t = \mathbf{f}_t^T \mathbf{W}^k, \quad \mathbf{V}_t = \mathbf{f}_t^T \mathbf{W}^v, \quad (7)$$

where $\mathbf{W}^q, \mathbf{W}^k, \mathbf{W}^v \in \mathbb{R}^{d_l \times d_l}$ are learnable projection matrices. The self-attention output is then computed as:

$$\text{Self-Attention} = \text{Softmax} \left(\frac{\mathbf{Q}_t \mathbf{K}_t^T}{\sqrt{d_l}} \right) \mathbf{V}_t = \mathbf{M}_t \mathbf{V}_t, \quad (8)$$

where $\mathbf{M}_t \in \mathbb{R}^{L \times L}$ denotes the self-attention map.

To enforce sparsity in $\mathbf{M}_t$, we introduce an entropy regularization term:

$$\mathcal{L}_{\text{sparse}} = - \sum_{i=1}^N \sum_{j=1}^N \mathbf{M}_t^{ij} \log(\mathbf{M}_t^{ij}), \quad (9)$$

where $\mathbf{M}_t^{ij}$ represents the attention weight between i -th and j -th sensor features across time steps. The entropy

loss applied to the self-attention map encourages sparsity by minimizing uncertainty in the attention distribution. As a result, the attention weights for less important or weakly related elements in the self-attention map approach zero, leading to sparsity, as shown in the bottom left in Fig. 2. In other words, the model concentrates on the relationships between important sensors while suppressing irrelevant sensor or noise-related connections, all without relying on any prior knowledge. Although it requires tuning the regularization weight to achieve optimal sparsity, this design allows the model to adaptively learn meaningful dependencies in a data-driven manner, enhancing both generalization and interpretability.

E. TOTAL LOSS FUNCTIONS

The total loss function in our framework integrates multiple objectives to ensure effective anomaly detection by leveraging both input space and latent space dynamics. It is composed of the following components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{input}} + \mathcal{L}_{\text{MRH}} + \lambda \mathcal{L}_{\text{sparse}}, \quad (10)$$

where λ is a hyperparameter that controls the relative weight of the sparse attention regularization term.

By balancing these loss terms, the total loss function ensures that the model learns to accurately reconstruct normal data, effectively distinguishes meaningful latent deviations, and prioritizes key sensor relationships. This combination improves the model's robustness and enhances its anomaly detection performance.

F. REFINED ANOMALY SCORE

In conventional reconstruction-based anomaly detection, anomalies are identified based on the input reconstruction error, which is the difference between the input data and its reconstruction. This is defined as the input anomaly score:

$$\text{Score}_{\text{input}} = \left[\|\mathbf{X}_t - \hat{\mathbf{X}}_t\|_2^2 \right]_{t=1, \dots, L}. \quad (11)$$

In our framework, DT-LA also utilizes the reconstruction error in the latent space to determine the latent anomaly score, which is defined as:

$$\text{Score}_{\text{latent}} = \left[\|\mathbf{z}_t - \hat{\mathbf{z}}_t\|_2^2 \right]_{t=1, \dots, L}. \quad (12)$$

Our goal is to refine the anomaly score by effectively combining these two criteria to identify anomalous time points robustly. In previous studies [8], [10] that incorporated additional metrics alongside input reconstruction errors, the common approach is to apply a softmax function to the additional metrics. The final anomaly score is then formulated by performing element-wise multiplication with the original reconstruction error. However, the softmax function normalizes the distribution based only on relative deviations within the window, which can lead to distortion in anomaly scores. For example, when all values within the window are relatively low (e.g., normal data), the softmax function tends to assign high anomaly scores to even

TABLE 2. Dataset statistics.

Dataset	Variables	Train	Valid	Test	p (%)
SMAP	25	108,146	27,037	427,617	1.0
MSL	55	46,653	11,664	73,729	1.0
PSM	26	105,984	26,497	87,841	1.0
SMD	38	566,724	141,681	708,420	0.5

minor fluctuations due to their relative deviations. Therefore, finding a balance between relative differences and absolute magnitudes is crucial for anomaly detection in multivariate time series, where both global trends and local deviations are important.

To address this issue, we propose an alternative strategy for refining the anomaly score by incorporating absolute value scaling. We introduce a scaled-softmax function, which preserves both the relative importance of each value and its original magnitude. This leads to a refined anomaly scoring function $AS(\cdot)$, defined as:

$$AS(\mathbf{X}_t) = \text{Score}_{\text{input}} \odot \text{Scaled-Softmax}(\text{Score}_{\text{latent}}), \quad (13)$$

where the scaled-softmax is formulated as:

$$\text{Scaled-Softmax}(s_i) = \frac{\exp(s_i)}{\sum_{j \in \text{window}} \exp(s_j)} \cdot |s_i|. \quad (14)$$

In this formulation, softmax emphasizes relative importance within the window, while scaling by the absolute magnitude of s_i preserves true error intensity. This prevents minor normal fluctuations from being overestimated, reducing false alarms. As a result, DT-LA captures meaningful latent deviations more accurately and produces more reliable anomaly scores, leading to improved detection performance.

IV. EXPERIMENTS

A. EXPERIMENTAL SETTINGS

1) IMPLEMENTATION DETAILS

We implement the proposed network using PyTorch [41] and train it on a single RTX 3090 GPU. The model is optimized using the Adam optimizer [42] with a learning rate of 10^{-5} . Training is conducted for a total of 100 epochs, with early stopping applied within the first 10 epochs. We use a batch size of 256. The Dual Transformers framework consists of 3 layers, each layer with a latent dimension $d_l = 512$ and 8 attention heads. The hyperparameter λ in (10) is set to 0.5 across all datasets.

2) DATASETS

We evaluate our method on four publicly available multivariate time series datasets with diverse domains and anomaly characteristics.

- **SMAP (Soil Moisture Active Passive satellite)** [43]: Data from NASA's Soil Moisture Active Passive satellite, consisting of 25 telemetry variables. Anomalies are often related to subsystem malfunctions and sensor glitches.

- **MSL (Mars Science Laboratory rover)** [43]: Collected from the Mars Science Laboratory rover, this dataset includes 55 telemetry variables and anomalies related to environmental changes or operational faults.
- **PSM (Pooled Server Metrics)** [44]: Metrics collected from application servers in a production e-commerce environment (eBay), including CPU, memory, and I/O usage. It has 26 variables and a mix of performance and configuration-related anomalies.
- **SMD (Server Machine Dataset)** [6]: Server machine metrics from a large internet company with 38 variables. The dataset includes multiple types of anomalies such as network congestion, memory leaks, and CPU overload.

The detailed statistics of the training, validation, and test time points for each dataset are presented in Table 2. A time point is labeled as anomalous if its final anomaly score, computed using (13), exceeds a predefined threshold. This threshold is automatically determined to ensure that $p\%$ of the validation dataset is labeled as anomalous, following prior works [8], [10].

3) PREPROCESSING

Following previous works [45], [46], we normalize all data using the mean and variance of the training dataset before proceeding. This normalization ensures that all features are scaled to have zero mean and unit variance, which is critical for stabilizing and accelerating the training process. We use a non-overlapping sliding window approach to extract sub-series, ensuring consistency across all datasets. The window size is fixed at 100 for fair comparisons with existing methods [8], [10].

4) EVALUATION METRICS

Anomalies in time series data often span multiple contiguous points, and detecting even one point within an anomalous segment should ideally count as a correct detection. To address this, we apply the point adjustment method [6], [47], [48], which is commonly used in time series anomaly detection. This method adjusts evaluation metrics to account for the temporal nature of anomalies. Specifically, if any point within an anomalous segment is detected as an anomaly, the entire segment is considered a true positive.

Using the point adjustment method, we evaluate our model with standard anomaly detection: precision, recall, and F1-score. These metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Here, TP, FP, and FN refer to true positives, false positives, and false negatives, respectively. By incorporating the point adjustment method, these metrics are adapted to better capture the context of anomalies in time series data, ensuring that our evaluation reflects the detection of anomalous segments rather than isolated points.

TABLE 3. Comparison with other methods on real-world multivariate time series datasets. Precision (P), recall (R), and F1-score (F1) are reported, where higher values indicate better performance. * denotes reproduced results using the same settings as our method, while other baseline performances are taken from [10]. The best performance is labeled in bold, and the second best is labeled in underlined.

Method	SMAP			MSL			PSM			SMD			Avg.
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	F1
OC-SVM	53.85	59.07	56.34	59.78	86.87	70.82	62.75	80.89	70.67	44.34	76.72	56.19	63.51
IsolationForest	52.39	59.07	55.53	53.94	86.54	66.45	76.09	92.45	83.48	42.31	73.29	53.64	64.78
LOF	58.93	56.33	57.60	47.72	85.25	61.18	57.89	90.49	70.61	56.34	39.86	46.68	59.02
DAGMM	86.45	56.73	68.51	89.60	63.93	74.62	93.49	70.03	80.08	67.30	49.89	57.30	70.13
MPPACD	88.61	75.84	81.73	81.42	61.31	69.95	76.26	78.35	77.29	71.20	79.28	75.02	76.00
Deep-SVDD	89.93	56.02	69.04	91.92	76.63	83.58	95.41	86.49	90.73	78.54	79.67	79.10	80.61
THOC	92.06	89.34	90.68	88.45	90.97	89.69	88.14	90.99	89.54	79.76	90.95	84.99	88.73
LSTM-VAE	92.20	67.75	78.10	85.49	79.94	82.62	73.62	89.92	80.96	75.76	90.08	82.30	80.99
BeatGAN	92.38	55.85	69.61	89.75	85.42	87.53	90.30	93.84	92.04	72.90	84.09	78.10	81.82
OmniAnomaly	92.49	81.99	86.92	89.02	86.37	87.67	88.39	74.46	80.83	83.68	86.82	85.22	85.16
InterFusion	89.77	88.52	89.14	81.28	92.70	86.62	83.61	83.45	83.52	87.02	85.43	86.22	86.38
A.T.*	93.66	99.41	96.45	90.91	92.72	91.81	97.39	97.79	97.59	87.96	94.68	91.20	94.26
DCdetector*	<u>94.36</u>	<u>98.99</u>	<u>96.62</u>	92.54	<u>97.87</u>	95.13	96.63	98.07	97.35	88.11	84.83	86.44	93.89
MEMTO*	93.58	<u>99.54</u>	96.47	92.00	94.82	93.39	<u>97.43</u>	<u>99.21</u>	<u>98.31</u>	<u>88.35</u>	<u>98.10</u>	<u>92.97</u>	<u>95.29</u>
DT-LA (Ours)	94.39	99.79	97.02	<u>92.07</u>	98.33	<u>95.09</u>	97.49	99.37	98.42	89.81	98.56	93.98	96.13

B. MAIN RESULTS

1) BASELINES

We extensively compare our method against 14 models, including traditional machine learning baselines (OC-SVM [1], Isolation Forest [2]), density-based models (LOF [3], DAGMM [49], MPPACD [50]), clustering-based models (Deep-SVD [51], THOC [48]), reconstruction-based models (LSTM-VAE [5], BeatGAN [14], OmniAnomaly [6], InterFusion [7], AnomalyTransformer [8], MEMTO [10]), and representation learning-based models (DCdetector [37]). Among these, AnomalyTransformer [8] (denoted as A.T.), DCdetector [37], and MEMTO [10] are state-of-the-art Transformer-based models. For fair comparison, we report the reproduced results for these models under the same experimental settings as ours (marked with *).

2) COMPARISON WITH EXISTING METHODS

Table 3 presents the evaluation results of anomaly detection performance across four benchmark datasets: SMAP, MSL, PSM, and SMD. Each method is assessed using precision (P), recall (R), and F1-score (F1), where higher values indicate better performance. Our proposed DT-LA achieves the highest average F1-score of 96.13% across all datasets, surpassing the previous state-of-the-art model MEMTO, which attained 95.29%. Specifically, DT-LA achieves the best performance on SMAP (97.02%), PSM (98.42%), and SMD (93.98%), and ranks second on MSL (98.33%), demonstrating its consistent and robust performance across diverse scenarios.

Traditional methods such as OC-SVM and Isolation Forest exhibit limited effectiveness ($F1 < 65\%$), primarily due to their inability to capture temporal dependencies and complex inter-sensor relationships. Deep learning models like

LSTM-VAE, BeatGAN, and OmniAnomaly show noticeable improvements by incorporating sequential modeling, yet typically remain below 90% in F1-score, highlighting limitations in handling subtle anomalies and overgeneralization. In contrast, Transformer-based models, including AnomalyTransformer (A.T.), DCdetector, and MEMTO, achieve consistently strong results ($F1 > 90\%$), underscoring the strength of attention mechanisms in modeling long-range dependencies and global context. Nevertheless, DT-LA achieves further improvements by introducing three novel components, notably attaining the highest recall across all datasets—SMAP (+0.25%), MSL (+0.46%), PSM (+0.16%), and SMD (+1.01%)—which is critical for identifying subtle or early-stage anomalies. At the same time, it maintains high precision by effectively suppressing false positives through its attention and scoring strategies. This balanced performance in both recall and precision demonstrates that DT-LA not only achieves high detection accuracy but also ensures strong generalization and reliability, making it highly suitable for real-world anomaly detection deployments.

C. ABLATION STUDY

To validate the effectiveness of each component in the proposed DT-LA framework, we conduct comprehensive ablation studies across four benchmark datasets. The analysis is organized into the following four parts: (1) loss analysis – the impact and interaction of reconstruction loss functions and sparse self-attention regularization, through both training dynamics analysis and ablation experiments; (2) effect of threshold δ in MRH Loss – examining the sensitivity of detection performance to the hyperparameter δ ; (3) effect of anomaly scoring criteria – comparing different strategies for scoring anomalies based on reconstruction

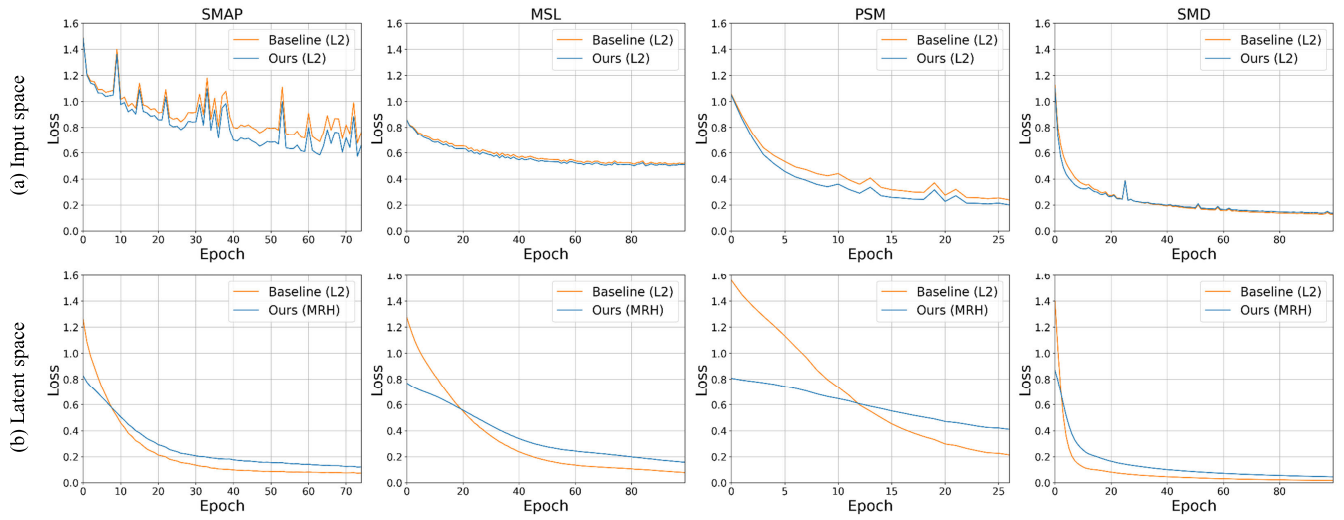


FIGURE 4. Training dynamics of reconstruction losses in (a) input and (b) latent space across four datasets: SMAP, MSL, PSM, and SMD. Each plot compares DT-LA (Ours) and the Baseline model. The Baseline uses L2 loss for both spaces, while DT-LA uses L2 in the input space and MRH in the latent space.

TABLE 4. Ablation studies on the types of latent space reconstruction loss functions and whether sparse attention is applied.

Latent Space	Sparse Attention	F1-score				
		SMAP	MSL	PSM	SMD	Avg.
L2	✗	96.51	93.27	97.63	93.62	95.26
	✓	96.64	93.30	97.81	93.94	95.42
MRH	✗	96.62	93.39	98.42	93.79	95.55
	✓	97.02	95.09	98.42	93.98	96.13

errors; (4) effect of regularization coefficient λ in sparse attention – evaluating how varying the entropy-based sparsity weight influences attention behavior and detection accuracy.

1) LOSS ANALYSIS

To investigate how different loss functions influence training behavior and detection performance, we analyze the training dynamics and ablation results of each key component and their combinations. Fig. 4 illustrates the temporal evolution of the input and latent reconstruction losses during training for both the Baseline and our proposed DT-LA. In the input space, both models converge similarly, indicating that applying MRH loss in the latent space does not compromise input reconstruction. In contrast, a clear difference emerges in the latent space: while the Baseline rapidly minimizes the latent loss to near zero, DT-LA maintains a higher latent loss that decays more gradually. This behavior is a direct consequence of the MRH formulation, which reduces the penalty for large deviations and allows the model to preserve meaningful anomaly-related variations in the latent representation. This dynamic prevents overfitting and helps retain informative features for anomaly detection.

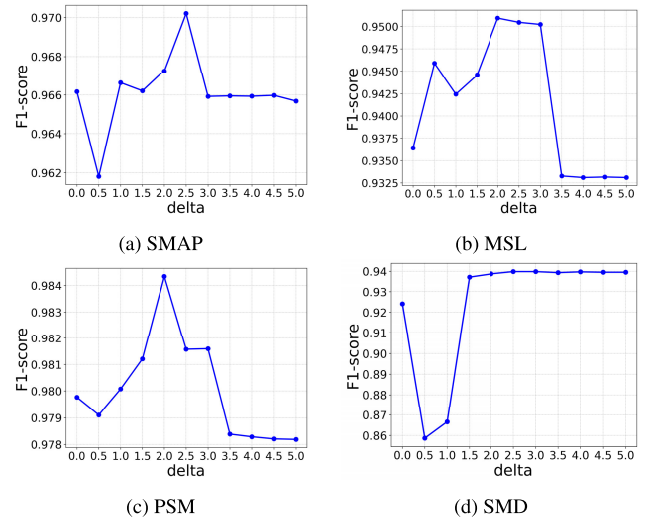


FIGURE 5. Impact of the threshold δ in Modified Reverse Huber loss on F1-scores.

Complementing this analysis, Table 4 presents ablation results comparing different configurations of latent loss functions (L2 vs. MRH) and sparse attention regularization (enabled or disabled). The results consistently show that replacing L2 with MRH in the latent space leads to noticeable improvements in F1-score across all datasets. Specifically, the average F1-score increases by 0.29% without sparse attention and by 0.71% with it. Furthermore, sparse self-attention provides additional gains in both settings: when combined with L2, it increases the F1-score from 95.26% to 95.42% (+0.16%), while with MRH, it improves performance from 95.55% to 96.13% (+0.58%). These results highlight two important findings: (1) MRH loss effectively mitigates overgeneralization by reducing penalties

TABLE 5. Ablation results on different anomaly scoring criteria.

Settings	Anomaly Criterion	SMAP			MSL			PSM			SMD			Avg.
		P	R	F1	P	R	F1	P	R	F1	P	R	F1	
(a)	S_{input}	92.53	57.68	71.06	89.45	86.93	88.17	99.63	85.77	92.18	69.86	95.15	80.06	82.87
(b)	S_{latent}	93.01	54.27	68.55	54.79	35.08	42.77	85.33	91.01	88.08	95.78	80.61	72.45	67.96
(c)	$S_{\text{input}} \odot \text{Softmax}(S_{\text{latent}})$	93.72	99.79	96.64	92.00	98.33	95.06	97.48	99.37	98.42	89.66	97.80	93.55	95.92
(d)	$S_{\text{input}} \odot \text{Scaled-Softmax}(S_{\text{latent}})$	94.39	99.79	97.02	92.07	98.33	95.09	97.49	99.37	98.42	89.81	98.56	93.98	96.13

for large latent deviations, and (2) sparse attention improves detection robustness by selectively modeling critical inter-variable dependencies. Notably, their combination yields the best performance, underscoring the complementary nature of these two components within the DT-LA framework.

2) EFFECT OF THRESHOLD δ IN MRH LOSS

The threshold δ in the MRH loss controls the model's sensitivity to latent reconstruction deviations and serves as a key hyperparameter. We determine its optimal value via grid search on the validation set of each dataset, selecting the one that maximizes F1-score. Despite relying on empirical tuning, an appropriately chosen δ consistently enhances anomaly separability across all datasets.

Fig. 5 shows the effect of varying δ across four benchmarks. In SMAP, MSL, and SMD, we observe a common trend: the F1-score increases as δ grows, reaches a clear peak, and then gradually declines to a level comparable to the baseline L2 loss. This saturation occurs because, at large δ values, the MRH loss becomes nearly equivalent to L2, losing its ability to emphasize informative deviations. These results empirically support the existence of an optimal dataset-specific δ that improves anomaly detection by suppressing overgeneralization. In PSM, although performance varies slightly, it also converges to baseline levels with large δ , reinforcing the overall trend. As a result, by reducing penalties on large deviations, MRH loss helps the model retain meaningful signals.

3) EFFECT OF ANOMALY SCORING CRITERIA

Table 5 presents the results of an ablation study on different anomaly scoring criteria in DT-LA. The results emphasize the significance of incorporating both input and latent-space deviations, as well as the efficacy of the proposed scaled-softmax normalization. When only the input anomaly scores are used (Setting (a)), the model achieves a moderate average F1-score of 82.87%. In contrast, relying solely on latent-space deviation (Setting (b)) results in a lower performance of 67.96% in the average F1-score. These results suggest that while latent anomaly scores provide valuable insights, the application of MRH loss, which partially limits the latent reconstruction capability, diminishes the anomaly detection performance when relying on latent anomaly scores alone. By integrating both input and latent anomaly scores using standard softmax

TABLE 6. Ablation results for the regularization coefficient λ in the entropy-based sparse attention loss.

λ	F1-score				
	SMAP	MSL	PSM	SMD	Avg.
0.1	96.44	93.42	98.2	93.68	95.43
0.3	96.45	93.35	98.21	93.76	95.44
0.5	97.02	95.09	98.42	93.98	96.13
0.7	96.91	95.04	98.42	93.76	95.98
1	96.82	94.93	98.22	93.82	95.94

normalization (Setting (c)), the performance improves significantly, yielding an average F1-score of 94.92%. This demonstrates that latent anomaly scores complement input anomaly scores, enhancing the model's anomaly detection capability. The best performance is achieved with the scaled-softmax normalization (Setting (d)), which results in an F1-score of 97.02%. This confirms that scaled-softmax effectively mitigates softmax-induced distortion, particularly in challenging scenarios where anomaly scores within a window are relatively low. The scaled-softmax function maintains a strong balance between precision and recall, reducing false positives while ensuring comprehensive anomaly detection. Overall, these results demonstrate that combining both input and latent anomaly scores improves performance, with scaled-softmax refining the scoring process and boosting robustness across datasets.

4) EFFECT OF REGULARIZATION COEFFICIENT λ IN SPARSE ATTENTION

To determine an appropriate value for the regularization coefficient λ in the sparse self-attention loss in (10), we conducted an ablation study over $\lambda \in 0.1, 0.3, 0.5, 0.7, 1.0$. As shown in Table 6, $\lambda = 0.5$ achieves the highest average F1-score across datasets, with a particularly strong effect on SMAP. This value provides a favorable balance between encouraging sparsity and preserving reconstruction quality. Smaller values lead to under-regularization and reduced anomaly sensitivity, while larger values result in excessive sparsification and degraded performance.

D. VISUAL ANALYSIS

To qualitatively evaluate our model under real-world conditions, Fig. 6 presents a comparison of anomaly scores

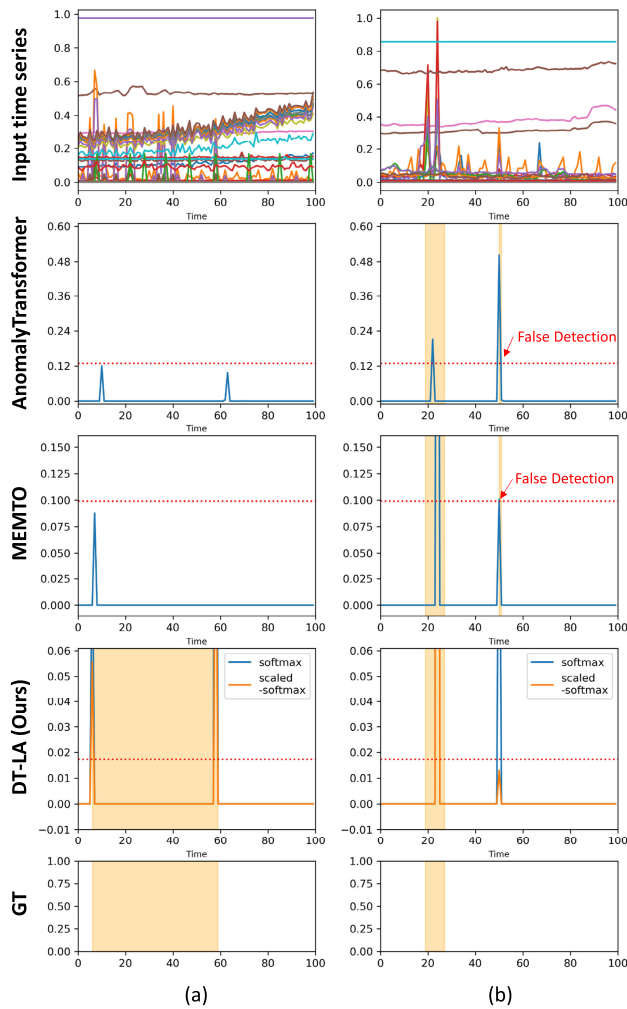


FIGURE 6. Visualization of the SMD dataset and its corresponding anomaly scores for each method.

produced by different methods on the SMD dataset. The top row shows the multivariate input time series, followed by the anomaly scores generated by AnomalyTransformer [8], MEMTO [10], and our proposed DT-LA. The red dotted line indicates the anomaly detection threshold, while shaded regions highlight time segments detected as anomalous. The bottom row displays the ground truth anomaly periods.

Panel (a) shows a case where our model detects a subtle anomaly that goes unnoticed by other methods. Both AnomalyTransformer and MEMTO yield low scores that remain below the threshold, failing to recognize the anomaly. In contrast, DT-LA effectively captures the deviation early and aligns closely with the ground truth. This improvement is attributed to the MRH loss, which amplifies informative deviations in the latent space, particularly those that conventional L2-based losses tend to ignore. On the other hand, panel (b) highlights a scenario prone to false alarms. While AnomalyTransformer and MEMTO incorrectly raise anomaly scores in normal regions, DT-LA suppresses such noise and activates only within the actual anomaly interval. This robustness

TABLE 7. Comparison of average trainable parameters (in MB) and training/inference time (in seconds) across four datasets.

Method	Trainable parameters (MB)	Training time (s)	Inference time (s)
AnomalyTransformer	18.42	297.36	17.55
DCdetector	3.36	670.88	72.53
MEMTO	22.52	99.1	7.21
DT-LA (Ours)	37.68	50.81	2.97

stems from our scaled-softmax anomaly scoring strategy, which balances absolute and relative deviations to reduce over-sensitivity to benign fluctuations. Together, these two cases demonstrate the core strengths of DT-LA: its heightened sensitivity to subtle but meaningful anomalies, and its ability to avoid false positives—both of which are essential for reliable anomaly detection in complex multivariate time series.

E. COMPUTATIONAL COSTS AND CONVERGENCE TIME

To evaluate the computational efficiency of transformer-based models, Table 7 compares the average number of trainable parameters, training time, and inference time across four datasets. Our DT-LA achieves the fastest training time (50.81 seconds) and inference time (2.97 seconds) with only a slight increase in the number of parameters, demonstrating its efficiency for real-world applications. Specifically, AnomalyTransformer and DCdetector incur significant computational overhead due to their distribution estimation processes to calculate sensor associations and multi-patch operations, resulting in longer training and anomaly score computation times. While MEMTO has made some improvements in training and inference times, it still requires a relatively long training process due to its two-stage learning approach, which involves storing normal data prototypes in memory. In contrast, DT-LA ensures efficient training through a streamlined reconstruction process, where both input and latent space reconstructions are performed end-to-end. The sparse self-attention mechanism allows the model to focus on key sensor dependencies, accelerating convergence. Additionally, DT-LA achieves the fastest inference time due to the simplicity of its anomaly scoring mechanism, making it particularly well-suited for deployment in large-scale time-series anomaly detection tasks.

V. CONCLUSION

In this paper, we present Dual Transformers with Latent Amplification (DT-LA), a novel framework designed to mitigate the overgeneralization problem in multivariate time series anomaly detection. DT-LA jointly leverages reconstruction in both input and latent spaces to enhance anomaly separability. To amplify informative deviations while avoiding excessive penalization of large errors, we introduce the Modified Reverse Huber (MRH) loss, which

emphasizes latent discrepancies indicative of anomalies. We further incorporate a sparse self-attention mechanism with entropy-based regularization to capture essential inter-sensor dependencies and reduce the influence of spurious correlations. Additionally, we propose a refined anomaly scoring strategy based on scaled-softmax, which balances absolute and relative deviations to yield stable and discriminative anomaly scores. Extensive experiments across four benchmark datasets demonstrate that DT-LA consistently outperforms state-of-the-art methods in both detection accuracy and computational efficiency.

Although DT-LA demonstrates strong performance, it also presents opportunities for further improvement. In particular, the effectiveness of the MRH loss depends on the appropriate setting of its hyperparameters, which currently requires empirical tuning for each dataset. Future work could explore adaptive or automated strategies to alleviate this dependency and further improve generalizability. Nevertheless, DT-LA's single-stage, memory-free architecture, coupled with its strong and balanced performance, makes it a practical and scalable solution for real-time anomaly detection in diverse application scenarios.

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YEJI CHOI received the B.S. and M.S. degrees in mathematics from Ewha Womans University, Seoul, South Korea, in 2014 and 2016, respectively. She is currently pursuing the Ph.D. degree in electrical and electronic engineering with Yonsei University, Seoul. She is a Student Researcher with the Artificial Intelligence and Robotics Institute, Korea Institute of Science and Technology, Seoul. Her research interests include computer vision, image synthesis, and recognition.



KWANGHOON SOHN (Senior Member, IEEE) received the B.E. degree in electronic engineering from Yonsei University, Seoul, South Korea, in 1983, the M.S.E.E. degree in electrical engineering from the University of Minnesota, Minneapolis, MN, USA, in 1985, and the Ph.D. degree in electrical and computer engineering from North Carolina State University, Raleigh, NC, USA, in 1992. He was a Senior Member of the Research Engineer with the Satellite

Communication Division, Electronics and Telecommunications Research Institute, Daejeon, South Korea, from 1992 to 1993, and a Postdoctoral Fellow with the MRI Center, Medical School of Georgetown University, Washington, DC, USA, in 1994. He was a Visiting Professor with Nanyang Technological University, Singapore, from 2002 to 2003. He is currently an Underwood Distinguished Professor with the School of Electrical and Electronic Engineering, Yonsei University. His research interests include 3D image processing and computer vision.



IG-JAE KIM (Member, IEEE) received the B.S. and M.S. degrees from the Department of EE, Yonsei University, Seoul, South Korea, in 1996 and 1998, respectively, and the Ph.D. degree from the Department of EECS, Seoul National University, in 2009. He was a Postdoctoral Researcher at the Media Laboratory, Massachusetts Institute of Technology (MIT), from 2009 to 2010. He joined Korea Institute of Science and Technology (KIST), Seoul, in 1998, and served as the Director for

the Center for Imaging Media Research, KIST, from 2017 to 2020. He is currently the Director-General of the AI and Robotics Institute, KIST. He is also an Associate Professor at Korea University of Science and Technology and an Adjunct Professor at Korea University. He has published over 100 fully-refereed papers in international journals and conferences, including *ACM Transactions on Graphics*, *IEEE TRANSACTIONS ON IMAGE PROCESSING*, *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*, *Pattern Recognition*, *SIGGRAPH*, *Eurographics*, *CVPR*, and *AAAI*. His research interests include pattern recognition, computer vision and graphics, deep learning, and computational photography.

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